METRIC
MIL-D-89001
26 February 1990
SUPERSEDING
PS/1CF/200
April 1986

MILITARY SPECIFICATION

DIGITAL TERRAIN ELEVATION DATA (DTED) LEVEL 2 (1CF)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification defines the requirements within Defense Mapping Agency's (DMA) Digital Terrain Elevation Data Base which supports various weapon and training systems.

1.2 Purpose. The purpose of this specification is to assure uniformity of treatment among all mapping and charting elements engaged in a coordinated production and maintenance program for this product.

1.3 Security. This specification and its resultant products are UNCLASSIFIED.

2. APPLICABLE DOCUMENTS

2.1 Government documents .

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the current Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Military Standards
MIL-STD-600001 MC&G Accuracy Standard

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, Defense Mapping Agency, ATTN: PR, 8613 Lee Highway, Fairfax, VA 22031-2137 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A **AREA MCGT**

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

2.1.2 Other Government documents, drawings, and publications.

This section is not applicable to this specification.

2.2 Non-Government publications.

This section is not applicable to this specification.

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Accuracy.

3.1.1 Absolute horizontal accuracy.

90% Circular Error (C.E.) World Geodetic System (WGS)
≤ 50 meters

3.1.2 Absolute vertical accuracy.

90% Linear Error (L.E.) Mean Sea Level (MSL)
≤ 30 meters

3.1.3 Relative horizontal accuracy (point-to-point).

90% C.E. WGS
≤30 meters over a 1 degree cell

3.1.4 Relative vertical accuracy (point-to-point).

90% L.E. MSL
≤20 meters over a 1 degree cell

Note: Reference MIL-STD-600001, for accuracy definitions.

3.2 Datum.

3.2.1 Horizontal datum. Horizontal Datum shall be the World Geodetic System (WGS). DTED shall be compiled to the current WGS datum or revised to be compatible with the current WGS datum.

3.2.2 Vertical datum. Vertical Datum shall be MSL.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

3.3 Product format. A data file of DTED is a 1° by 1° cell defined by whole degree latitude and longitude lines on WGS. A data file of DTED Level 2 may be less than a full 1° cell and can be increments of 15' x 15'. A DTED data file shall not cross whole degree latitude or longitude lines.

3.3.1 Adjacent data files. Adjacent data files shall not have gaps between them and the only overlap that exists is along adjacent boundaries. All adjacent boundaries shall be coincident.

3.4 Standard format. The DMA Standard Terrain Format is DMA's standardized system of recording terrain elevation data on magnetic tape. The format is intended for the purpose of production, storage and exchange of terrain elevation data.

3.4.1 Physical characteristics of magnetic tape. The physical characteristics of magnetic tape are as follows:

a. Length: 2400 feet

b. Width: .5 inch

c. Nine track recording format.

d. Odd parity.

e. Inter-record gap: .6 inch (6250 Frames Per Inch (FPI): .3 inch).

f. Physical end-of-tape markers at the beginning (beginning-of-tape marker) and end of the tape (end-of-tape marker).

g. Density-recording methods.

(1) 1600 FPI/Phase encoded. This is preferred by DMA for storage and data exchange and will normally be expected from producers and provided to requestors.

(2) 800 FPI/Non Return To Zero Indicator (NRZI). This is not preferred by DMA but shall be provided to or accepted from data requestors or producers unable to accept or generate 1600 FPI/phase encoded data.

(3) 6250 FPI/Group Coded Recording (GCR). Future preferred DMA data exchange format (shall not be used unless agreed to by sender and receiver).

3.4.2 Type of Input/Output (I/O). The individual records contain no system dependent generated control words.

3.5 Record characteristics

3.5.1 Recorded labels. American National Standard Magnetic Tape Labels for Information Interchange X3.27-1978. Recorded in ASCII code.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

3.5.2 Data records.

a. Record size. Variable length, maximum 7214 frames, minimum 14 frames, modal (average) 2414 frames.

b. Blocking factor. 1:1 (block size = record size).

c. Record sequence.

```
VOL      1     (Volume Header Label)
HDR      1     (Header Label for file A)
UHL      1     (User Header Label for file A)
*
DSI            (Data Set Identification Record for file A)
ACC            (Accuracy Description Record for file A)
DATA           (for file A)
*
EOF      1     (End of File for file A)
UTL      1     (User Trailer Label for file A)
*
HDR      1     (Header Label for file B)
UHL      1     (User Header Label for file B)
*
DSI            (Data Set Identification Record for file B)
ACC            (Accuracy Description Record for file B)
DATA           (for file B)
*
EOF      1     (End of File for file B)
UTL      1     (User Trailer Label for file B)
*
*
```

Note: In the above sequence a Tape mark (hardware end of file) is denoted by an "*".

3.6 File description. A data file of DTED is a cell defined by latitudes and longitudes of a geographic reference system. The terrain elevation information is expressed in meters. The elevation posts are defined by the intersections of rows and columns within a matrix. The required matrix intervals, defined in terms of geographic arc seconds, vary according to latitude (see table I).

Note: All values in seconds are in terms of arc measure.

3.7 Logical characteristics.

3.7.1 Data file structure. Arranged in 1 degree by 1 degree geographic areas. Each data file shall contain data falling within a single one degree cell. The reference origin for each data file shall be the Southwest corner of the degree cell. Multiple data files shall be arranged primarily by ascending latitude bands (90 degrees South to 89 degrees North), secondarily by ascending longitude (180 degrees West to 179 degrees East). 1° cells may contain multiple data files in increments of 15' x 15'.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE I. Matrix intervals.

ZONE	LATITUDE	MATRIX INTERVAL	
		latitude	longitude
I	0° - 50° North-South	1 X	1 seconds
II	50° - 70° North-South	1 X	2 seconds
III	70° - 75° North-South	1 X	3 seconds
IV	75° - 80° North-South	1 X	4 seconds
V	80° - 90° North-South	1 X	6 seconds

3.7.2 Files extent. To provide overlap between adjacent data files, the degree cell coverage in this standard includes the integer degree values on all sides of the area. Each data record has one point of overlap with the cell above and one with the cell below (if the record extends to the degree cell limits). Entire data records lying on integer degree longitude values shall exist also in the adjacent degree cell. Data files will not cross integer degree latitude or longitude lines. Adjacent data files shall not have gaps between them and the only overlap that exists is along adjacent boundaries. All data files derived from coincident boundaries of adjacent cells shall be comprised of duplicate data records.

3.7.3 Terrain elevation intervals . The horizontal plane spacing of the elevation array will be in whole second intervals. (See table I.)

3.7.4 Data value sequence. The elevations within a data record have a constant longitude value. The first data value is the southernmost known elevation and the last is the northernmost. No two data records shall have the same longitude value.

3.7.5 Data record sequence . Within a data file, the data records are arranged by ascending (west to east) longitude.

3.7.6 Hash control total information. The last four frames of each data record contains a 32 bit value which is a checksum computed algebraically by summing all elevations and header words in that block, as 8 bit values using integer arithmetic. Each tape frame is considered as an unsigned 8-bit value for checksum calculation.

3.7.7 File structure diagram . (See tables II and III.)

3.8 Data record characteristics .

3.8.1 Numeric value. All elevation values are signed magnitude binary integers, right justified, 16 bits (2 bytes). The sign is the high order position. Negative values are not complemented.

3.8.2 Elevation values. The numeric value identified in paragraph 3.8.1 will allow a theoretical range of elevations between ± 32,767 meters; however in practice, the terrain elevation values shall not exceed + 9,000 meters or - 12,000 meters.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE II. File structure diagram-Terrain example, (1 of 2).

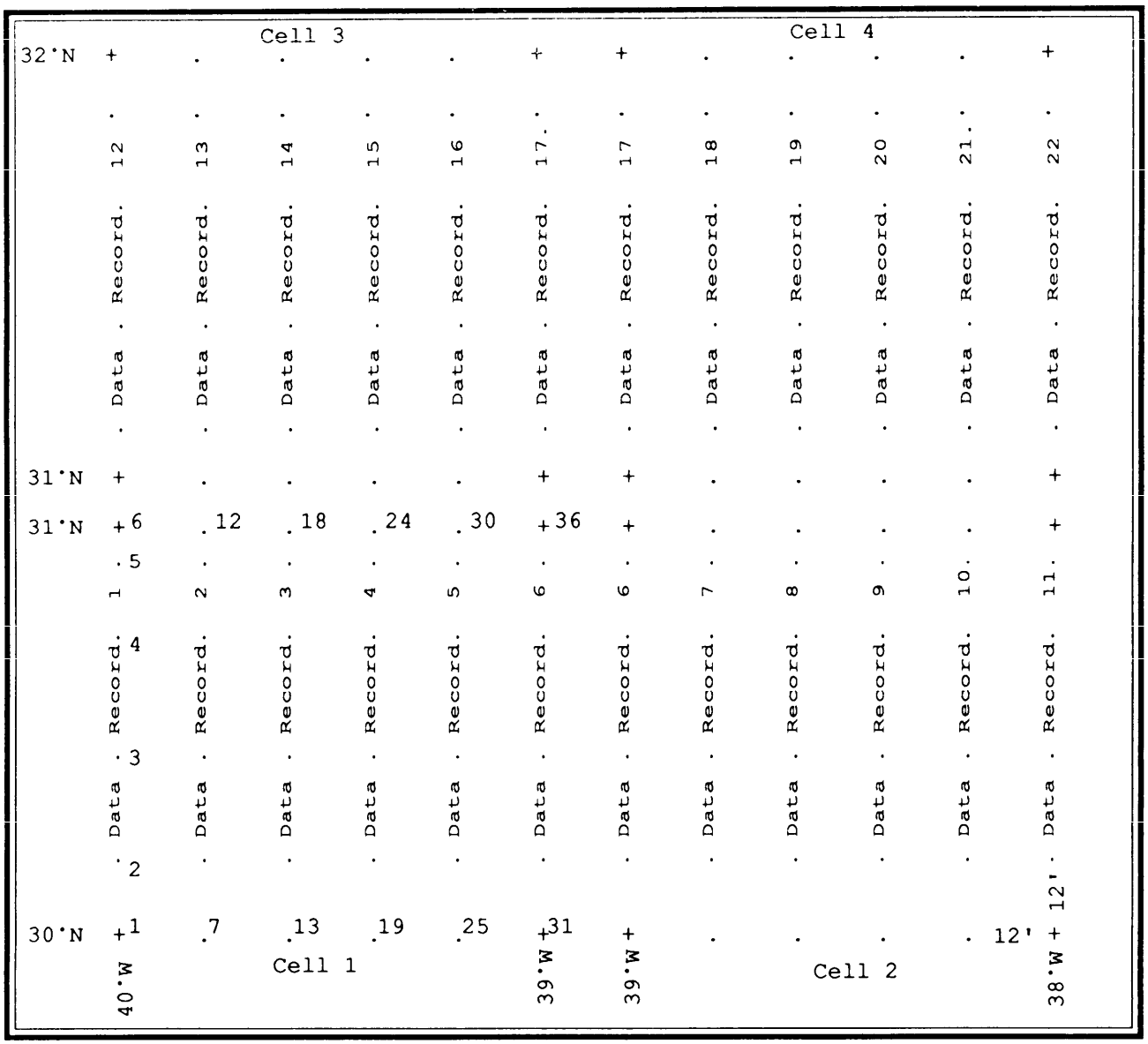
FOUR 1° CELLS 12' LONGITUDE SPACING (NON STANDARD)			
DATA STRUCTURE SEQUENCE	DATA STRUCTURE TYPE	DATA STRUCTURE SEQUENCE	DATA STRUCTURE TYPE
1	VOL 1	32	HDR 1
2	HDR 1	33	UHL 1 31°N, 40°W
3	UHL 1 30°N, 40°W	34	*
4	*	35	DSI
5	DSI	36	ACC
6	ACC	37	Data Record 12
7	Data Record 1	38	Data Record 13
8	Data Record 2	39	Data Record 14
9	Data Record 3	40	Data Record 15
10	Data Record 4	41	Data Record 16
11	Data Record 5	42	Data Record 17
12	Data Record 6	43	*
13	*	44	EOF 1
14	EOF 1	45	UTL 1
15	UTL 1	46	*
16	*	47	HDR 1
17	HDR 1	48	UHL 1 31°N, 39°W
18	UHL 1 30°N, 39°W	49	*
19	*	50	DSI
20	DSI	51	ACC
21	ACC	52	Data Record 17
22	Data Record 6	53	Data Record 18
23	Data Record 7	54	Data Record 19
24	Data Record 8	55	Data Record 20
25	Data Record 9	56	Data Record 21
26	Data Record 10	57	Data Record 22
27	Data Record 11	58	*
28	*	59	EOF 1
29	EOF 1	60	UTL 1
30	UTL 1	61	*
31	*	62	*

Note: * = Tape Mark

3.8.3 Null elevations. Unknown elevation data shall be represented by all "one" bits when the post is between known elevation data within the structure of a data record. This null elevation value of - 32,767 meters is used as a place holder in the data record. Null elevation values are allowed in 1 degree cells (i.e., partial cells) which have not been fully compiled. Null elevation values shall not be contained in 1 degree cells (i.e., full cells) issued from DMA to users as a standard product. Multiple elevation posts could occur in a 1° cell because Level 2 may be stored in 15' x 15' non-contiguous blocks.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE III. File structure diagram-Terrain example, 2 of 2.



Notes: (1) In the above example (non-standard 12' latitude/longitude spacing), each Data Record contains 6 elevations. A 1" standard 1° cell Data Record contains 3601 elevations.

(2) Elevations along 1° boundaries are repeated for each 1° cell.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the currentversion before use.

3.8.4 Missing Elevations.

a. Elevation posts which are missing prior to the first known elevation are not represented. The location of the first known point is given by the latitude count field in the Data Record.

b. Elevation posts which are missing past the last known elevation are not represented. The last known elevation is immediately followed by a checksum.

c. Missing elevations are allowed in 1 degree cells (partial cells) which have not been fully compiled.

d. Missing elevations are not contained in full 1 degree cells issued from DMA to users as a standard product.

3.9 Description of record contents. In the following record formats, an ASCII character requires one byte or 8 binary bits.

a. Volume Header Label (VOL).

Field Contents	Field Length In Characters	Character Start	Description
VOL	3	1	Recognition sentinel.
1	1	4	Fixed by standard.
	6	5	Reel Number. Six alphanumeric characters identifying the physical reel. (Requires leading alpha and trailing numeric.)
Blank or Nonblank	1	11	*Nonblank indicates restricted access, as the tape reel is privately owned.
Blanks	26	12	Unrequired available space.
Account Number	14	38	*Account number of owner of this tape reel. (DMA uses a maximum of 12 characters left-justified, space filled.)
Blanks	28	52	Fixed by standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

1	1	80	Fixed by standard.

Note: *These fields, to be defined by the producer, may be left blank.

b. File Header Label (HDR).

Field Contents	Field Length In Characters	Character Start	Description
HDR	3	1	Recognition sentinel.
1	1	4	Fixed by standard.
Filename	17	5	*Left-justified filename The first 12 characters are referenced by the Executive System for comparison with the filename portion of the external filename.
UNIVAC	6	22	*Fixed as set identifier when referenced by system.
0001	4	28	*Reel sequence number within a file.
0001 - NNNN	4	32	*File sequence number within a reel.
0001 00	4 2	36 40	*Generation and version numbers which are fixed at 1 and 0.
ØYYDDD	6	42	Creation date of tape. (A blank followed by two characters for the year, followed by three characters for the day (001 through 366) within the year date tape was written.)
ØYYDDD	6	48	*Expiration date of tape. Same format as creation date field. The date after which this tape reel may be considered as available for reallocation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

Field Contents	Field Length In Characters	Character Start	Description
A space indicates unlimited access to this reel. 15_8 - This reel is catalogued (on tape). 35_8- This reel is catalogued with read key. 55_8 - This reel is catalogued with write key. 75_8 - This reel is catalogued with read and write key.	1	54	*Accessibility.
Block Count	6	55	*Fixed at zeros.
Qualifier	13	61	*Used by the Executive Operating System.(DMA uses a maximum of 12 characters left-justified space filled.)
Blanks	7	74	Fixed by standard.

Note: *These fields, to be defined by the producer, may be left blank.

c. User Header Label (UHL).

Field Contents	Field Length In Characters	Character Start	Description
UHL	3	1	Recognition sentinel.
1	1	4	Fixed by standard.
DDDMMSSH	8	5	Longitude of origin (lower left corner of data set; Level 2: full degree value; leading zero(s) for all subfields: degrees, minutes and seconds). H is the Hemisphere of the data.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

Field Contents	Field Length In Characters	Character Start	Description
DDDMMSSH	8	13	Latitude of origin (lower left corner of data set; Level 2: full degree value; leading zero(s) for all subfields: (degrees, minutes and seconds). H is the Hemisphere of the data.
SSSS	4	21	Longitude data interval in seconds (decimal point is implied after third integer).
SSSS	4	25	Latitude data interval in seconds (decimal point is implied after third integer).
0000-9999 or NA	4	29	Absolute Vertical Accuracy in Meters. (With 90% assurance that the linear errors will not exceed this value relative to mean sea level (Right justified)).
T - Top Secret S - Secret C - Confidential U - Unclassified R - Restricted	3	33	Security Code. (Left justified)
Unique Reference	12	36	*Unique reference number (Provides Number pointer to file containing detailed file description).
Number of longitude lines	4	48	Count of the number of longitude (profiles) lines.
Number of latitude points	4	52	*Count of the number of latitude points per longitude line.
Multiple accuracy	1	56	0 - Single 1 - Multiple

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

Field Contents	Field Length In Characters	Character Start	Description
Reserved	24	57	Unused portion for future use.

Note: *These fields, to be defined by producer, may be left blank.

d. Data Set Identification (DSI) Record.
Fixed Length = 648 ASCII Characters
1 ASCII Character = 1 Byte = 8 Bits

Field Contents	Field Length In Characters	Character Start	Description
DSI	3	1	Recognition Sentinel.
T - Top Secret S - Secret C - Confidential U - Unclassified R - Restricted	1	4	Security Classification Code.
	2	5	Security Control and Release Markings. For DoD use only (DIAM 65-19).
	27	7	Security Handling Description. Other security description. (Free text or blank filled).
	26	34	Reserved for future use. (Blank filled).
DTED2	5	60	DMA Series Designator for product level.
	15	65	Unique reference number. (For producing nations own use (free text or zero filled)).
	8	80	Reserved for future use. (Blank filled).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

Field Contents	Field Length In Characters	Character Start	Description
01-99	2	88	Data Edition Number.
A-Z	1	90	Match/Merge Version.
YYMM	4	91	Maintenance Date. (Zero filled until used.)
YYMM	4	95	Match/Merge Date. (Zero filled until used.)
0000	4	99	Maintenance Description Code. (All zero filled.)
CCAAABBB (Country - Free Text)	8	103	Producer Code. (DIA Country Codes used for first 2 characters).
	16	111	Reserved for future use. (Blank filled).
SPECDTED2	9	127	Product Specification Stock Number
00-99	2	136	Product Specification Amendment and Change Number.
YYMM	4	138	Product Specification Date (Currently 9002.)
MSL	3	142	Vertical Datum (Mean Sea Level).
WGS84	5	145	Horizontal Datum Code (Current Version World Geodetic System).
	10	150	Digitizing Collection System. (Free text).
YYMM	4	160	Compilation Date. (Most descriptive month/year).
	22	164	Reserved for future use. (Blank filled).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

Field Contents	Field Length In Characters	Character Start	Description
DDMMSS.SH	9	186	Latitude of origin of data– leading zero for values less than 10; H is the hemisphere of the data.
DDDMMSS.SH	10	195	Longitude of origin of data– leading zero for values less than 100; H is the hemisphere of the data.
DDMMSSH	7	205	Latitude of SW corner of data, bounding rectangle– leading zero for values less than 10; H is the hemisphere of the data.
DDDMMSSH	8	212	Longitude of SW corner of data, bounding rectangle– leading zero for values less than 100; H is the hemisphere of the data.
DDMMSSH	7	220	Latitude of NW corner of data, bounding rectangle– leading zero for values less than 10; H is the hemisphere of the data.
DDDMMSSH	8	227	Longitude of NW corner of data, bounding rectangle– leading zero for values less than 100; H is the hemisphere of the data.
DDMMSSH	7	235	Latitude of NE corner of data, bounding rectangle– leading zero for values less than 10; H is the hemisphere of the data.
DDDMMSSH	8	242	Longitude of NE corner of data, bounding rectangle– leading zero for values less than 100; H is the hemisphere of the data.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

Field Contents	Field Length In Characters	Character Start	Description
DDMMSSH	7	250	Latitude of SE corner of data, bounding rectangle- leading zero for values less than 10; H is the hemisphere of the data.
DDDMMSSH	8	257	Longitude of SE corner of data, bounding rectangle- leading zero for values less than 100; H is the hemisphere of the data.
DDDMMSS.S	9	265	Clockwise orientation angle of data with respect to true North. (Will usually be all zeros for DTED.)
SSSS	4	274	Latitude interval in tenths of seconds between rows of elevation values. (Decimal point is implied after third integer.)
SSSS	4	278	Longitude interval in tenths of seconds between columns of elevation values. (Decimal point is implied after third integer.)
0000-9999	4	282	Number of Latitude lines, Actual count - Number of latitude points (rows that contain data).
0000-9999	4	286	Number of Longitude lines, Actual count - Number of longitude points (columns that contain data).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

Field Contents	Field Length In Characters	Character Start	Description
00 or 01-99	2	290	Partial Cell Indicator 00 = Complete 1° cell 01-99 = % of coverage completed.
	101	292	Reserved for DMA use only. (Blank filled.)
	100	393	Reserved for producing nation use only. (Free text or blank filled.)
	156	493	Reserved for future use. (Blank filled.)

e. Accuracy Description (ACC) Record.
Fixed Length = 2700 ASCII Characters
1 ASCII Character = 1 Byte = 8 Bits

Field Contents	Field Length In Characters	Character Start	Description
ACC	3	1	Recognition Sentinel.
0000-9999 or NA	4	4	*Absolute Horizontal Accuracy of Product in meters.
0000-9999 or NA	4	8	*Absolute Vertical Accuracy of Product in meters.
0000-9999 or NA	4	12	*Point-to-Point Horizontal Accuracy of Product in meters.
0000-9999 or NA	4	16	*Point-to-Point Vertical Accuracy of Product in meters.
	4	20	Reserved for future use. (Blank filled.)
	1	24	Reserved for DMA use only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

Field Contents	Field Length In Characters	Character Start	Description
	31	25	Reserved for future use. (Blank filled.)
00 or 02-09	2	56	Multiple Accuracy Outline Flag. 00 = No accuracy subregions provided. 02-09 = Number of accuracy subregions per 1° cell (maximum 9).

Note: *If Product has subregional accuracies, the overall accuracy of the product will be the worst accuracy.

Start of Accuracy Subregion Description. Repeat to maximum of nine times. Blank fill all unused accuracy subregions or unused coordinate pairs within a subregion. (1 Subregion = 284 ASCII Characters). Refer to paragraph 3.10.5 for accuracy subregion description.

0000-9999 or NA	4	58	Absolute Horizontal Accuracy of subregion in meters.
0000-9999 or NA	4	62	Absolute Vertical Accuracy of subregion in meters.
0000-9999 or NA	4	66	Point-to-Point Horizontal Accuracy of subregion in meters.
0000-9999 or NA	4	70	Point-to-Point Vertical Accuracy of subregion in meters.
03-14	2	74	Number of coordinates in accuracy subregion outline. (Maximum of 14 coordinate pairs. Coordinates are input clockwise. Implied closing from last to first coordinate pairs.)

Start of Coordinate Pair Description. Repeat to maximum of fourteen times to outline subregion. Blank fill all unused accuracy subregions and unused portions of subregions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

Field Contents	Field Length In Characters	Character Start	Description
DDMMSS.SH	9		Latitude–leading zero for values less than 10; H is the hemisphere of the data.
DDDMMSS.SH	10		Longitude–leading zero for values less than 100; H is the hemisphere of the data.
	End Coordinate Pair Description		
	End Accuracy Subregion		Description
	18	2614	Reserved for DMA use only.
	69	2632	Reserved for future use.

f. File Trailer Label (EOF).

Field Contents	Field Length In Characters	Description
EOF	3	Recognition Sentinel.
1	1	Fixed by standard.

Note: (See HDR for remainder of EOF fields.)

g. User Trailer Label (UTL).

Field Contents	Field Length In Characters	Description
UTL	3	Recognition Sentinel.
1	1	Fixed by standard.

Note: (See UHL for remainder of UTL fields.)

h. Data Record Description. Each element is a true elevation referenced to mean sea level datum recorded to the nearest meter. The horizontal position is referenced to precise longitude-latitude locations in terms of the current World Geodetic System (WGS), determined for each file by reference to the origin at the Southwest corner. The elements are evenly spaced in latitude and longitude at the interval designated in the user header label in South to North profile sequence.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the current version before use.

Field Contents	Field Length In Bytes	Description
252_8	1	Recognition Sentinel.
Data block count	3	Sequential count of the block within the file, starting with zero for the first block (Fixed Binary).
Longitude count	2	Count of the meridian. True longitude = longitude count x data interval + origin (SW corner longitude) (Fixed Binary).
Latitude count	2	Count of the parallel. True latitude = latitude count x data interval + origin (SW corner latitude) (Fixed Binary).
Elevation 1	2	True elevation value of point 1 of meridian in meters (Fixed Binary).
Elevation 2	2	True elevation value of point 2 of meridian in meters (Fixed Binary).
Elevation N	2	True elevation value of point N of meridian in meters (Fixed Binary).
Checksum	4	Algebraic addition of contents of block. Sum is computed as an integer summation of 8-bit values (Fixed Binary).

Note: Fixed Binary denotes signed magnitude, right-justified binary integers.

3.10 Explanation of records and fields (DTED).

3.10.1 Volume header label (VOL). This record is required for labeled tapes in accordance with American National Standards Institute (ANSI) standard X3.27-1978, Magnetic Tape Labels for Information Interchange.

3.10.2 File header label (HDR). This record is required for labeled tapes in accordance with ANSI standard X3.27-1978, Magnetic Tape Labels for Information Interchange.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

3.10.3 User Header Label (UHL). ANSI standard allows an optional user header label in the first file of a labeled tape. Several computer manufacturers have implemented tape labeling in such a way that the user header label in the first file of the tape is inaccessible. This record is maintained for minimum impact to users not desiring to use the DSI record, but all information in it is in the DSI record as well.

3.10.3.1 Fields(UHL). The following are explanations of fields within the UHL.

a. Longitude of Origin - Origin is always a full degree value even though the format allows values to be expressed to the second.

b. Latitude of Origin - Origin is always a full degree value even though the format allows values to be expressed to the second.

c. Seconds Longitude Interval - A cell of DTED is North-South oriented with columns of elevation posts running from south to north. The longitude interval is the East-West distance between the columns expressed as tenths of seconds.

d. Seconds Latitude Interval - The spacing between the elevation posts within a column (i.e., the distance between the rows) is the latitude interval expressed in tenths of a second.

3.10.4 Data Set Identification (DSI) Record. This record provides identification and security information relating to the DTED. The record is fixed length consisting of 648 ASCII characters. Each character is represented by 1 tape byte or 8 bits. Certain fields in the DSI record are duplicated in the UHL/UTL. These fields are required to match. Users may process the data using only the information in the DSI record if desired.

Note: Numeric values shall be right justified with leading zeros. Alpha and alphanumeric characters shall be left justified unless 'free text' is indicated in description.

3.10.4.1 Fields(DSI).

a. Security control and release markings - This field may contain the two character codes from DIAM 65-19 or may be blank filled .

b. DMA series designator - Five character code identifying the product in DMA Area Requirements and Product Status (ARAPS) file. DTED2 - Level 2.

c. Unique reference number - Free text or zero filled.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

d. Data edition number - The number assigned to the data indicating either original compilation (Edition 1) or subsequent replacements of the data (Editions 2, 3, etc.) in the Data Base (DB) to achieve accuracy requirements (recompilation) or currency/specification requirements (revision).The data edition number does not reflect the number of replacements made to the data to effect boundary matches.

e. Match/merge version - The number of times an edition of the data was changed to effect boundary continuity with adjacent data in the DB. Alphabetic Code A - Z designates data change for boundary continuity.

f. Maintenance date - YYMM (year and month); The date existing DB data was either revised (updated) to meet the currency requirements (or to effect specification changes), or recompiled to meet accuracy requirements. When the existing data is only revised (horizontal position or vertical values are not significantly changed) the maintenance date will reflect the date of the revision, but the compilation date will not be changed. It will continue to reflect the date of the original compilation. However, when the data is subjected to a major recompilation, the Compilation Date and the Maintenance Date will both be changed to reflect the date of the recompilation.

g. Match/merge date - YYMM (year and month);The latest date the data in the DB was changed to effect continuity with adjacent data. This data corresponds to the Match/Merge Version Code.

h. Maintenance description code - Must be zero filled.

i. Producer code - The first two characters (left justified) indicate the producing nation and are from DIAM 65-18 - Geopolitical Elements and Related Files. The last six characters are to be used at the discretion of the producer. Blanks are acceptable.

Belgium	BE	Netherlands	NL
France	FR	Norway	NO
Germany, Federal Republic of	GE	United Kingdom	UK
Italy	IT	United States	US

j. Product specification stock number - Identifies the product specification containing the compilation and accuracy requirements used to produce the data. Currently SPECDTED2.

k. Product specification amendment and change number - Indicates the highest numbered amendment or change used to produce the data. The first character identifies the Amendment number and the second character identifies the Change Notice number. (Amendment 0, Change 1-01; Amendment 2, Change 2-22; etc.)

l. Vertical datum code - Currently MSL.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

m. Horizontal datum code - Identifies the version of WGS to which the product is horizontally controlled. During original compilation DTED uses the WGS datum then in effect for DMA use. Upon transition to a new WGS datum the existing DTED can be:

(1) Recompiled from an original source; and therefore carry a new Edition number.

(2) Recompiled by mathematically adjusting the elevation values; and therefore carry a new Edition number.

(3) Revised by defining the same elevations as compatible with the new WGS datum within the accuracy evaluations quoted; and therefore carry the same Edition number but with a new Maintenance Date.

n. Digitizing collection system - Identifies the equipment used to collect the elevation values from the source material used, i.e., AGDS, LIS, UNAMACE, AS11, etc.

o. Compilation date - YYMM (year and month); The date the data was either originally compiled (Edition 1) or the date existing DB data was subjected to a major recompilation which involved significant changes to the horizontal positions and vertical values. (Edition 2, 3, 4, etc.)

p. Latitude of origin - Expressed in degrees, minutes, seconds and tenths of seconds with N or S to indicate hemisphere.

q. Longitude of origin - Expressed in degrees, minutes, seconds and tenths of seconds with East or West to indicate hemisphere.

3.10.5 Accuracy description record (ACC). The accuracy record provides accuracy information relating to the DTED. The record is fixed length consisting of 2700 ASCII characters. Each character is represented by 1 tape byte or 8 bits. The record allows for up to nine accuracy subregions within the product should the accuracies of various portions of this product differ. Each subregion may consist of up to fourteen coordinate pairs to accurately portray the outline. Coordinates are input clockwise. Unused coordinate pairs are blank filled.

Note: Numeric values shall be right justified with leading zeros. Alpha and alphanumeric characters shall be left justified.

3.10.5.1 Accuracy subregion description . A subregion description contains the subregion accuracies, the number of subregion coordinate pairs and the actual coordinate pairs which define the subregion area. Unused subregions and/or unused coordinate pairs within a subregion will be blank filled. The subregions will be organized as follows:

a. The first coordinate pair of the first subregion must be the Southwest corner of the bounding rectangle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

b. The first coordinate pair of each subregion must be the most southern and western point in the subregion.

c. The first coordinate pair of each sequential subregion must be arranged in a clockwise direction starting with the first subregion. Subregions which contain at least part of a degree line will take precedence and be numbered clockwise, starting in the SW corner and proceeding around the four bounding degree lines. Any remaining sub-regions will be numbered in a south to north, west to east manner based on initial point within each subregion.

d. Coordinate pairs within a subregion must be arranged in a clockwise direction.

e. Each subregion coordinate pair must be located on or within the bounding rectangle.

f. Each coordinate pair in the bounding rectangle must be represented as a subregion coordinate pair.

g. Subregion areas must not overlap or leave gaps. The entire area defined by the bounding rectangle must be covered by subregion areas.

h. The first and last coordinate pairs of a subregion shall not be the same. Closure of the subregion area is implied.

i. A subregion can only be generalized if all of the subregion descriptions have been used.

j. Subregion format:

Subregion Number	Field Length in Characters	Character Start
1	284	58
2	284	342
3	284	626
4	284	910
5	284	1194
6	284	1478
7	284	1762
8	284	2046
9	284	2330

3.11 Feature requirements/inclusion condition .

a. Feature code: 9B045

(1) Category: General (9)
(2) Subcategory: Control Points (B)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the currentversion before use.

b. Feature: Elevation Point

c. Feature Definition: A point within a matrix of points having a precise position on the earth's surface with the elevation relative to MSL.

d. Feature Type: Point

e. Required Attribute: ZVL (Z Value).
The elevation of the feature as referenced to MSL.

(1) Increment: 1 meter

(2) Limits: From Mean Sea Level to the tallest portion of the feature.

(3) Variance: NA

(4) Range: -400...9998

f. Inclusion Condition: Precise 1 arc second horizontal position referenced to the origin of the southwest corner, evenly spaced in latitude and longitude in a South to North profile sequence.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in a contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use their own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure products and services conform to prescribed requirements.

4.1.1 Responsiblity for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsiblity of ensuring that all products or services submitted to the Government for acceptance comply with all requirements of a contract. Sampling inspection, as part of operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective products, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Final product quality. Final product quality will reflect the quality elements expressed by each applicable Military Standard.

5. PACKAGING

This section is not applicable to this specification.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification defines the content of Digital Terrain Elevation Data (DTED) which supports various weapon and training systems.

6.2 Supersession. This specification supersedes the Defense Mapping Agency Product Specifications for Digital Terrain Elevation Data (DTED), PS/1CF/200, Second Edition, April 1986.

6.3 Definitions.

6.3.1 Accuracy. The degree of conformity with which horizontal position and vertical values are represented on a map, chart, or related product in relation to an established standard.

a. Horizontal accuracy, absolute. The uncertainty in the horizontal position of a point with respect to the World Geodetic System caused by random and uncorrected systematic errors. The value is expressed as a circular error at the 90% confidence level.

b. Horizontal accuracy, point-to-point. The uncertainty in horizontal position between two points caused by random errors. The value is expressed as a circular error at the 90% confidence level.

c. Vertical accuracy, absolute. The uncertainty in the height of a point with respect to Mean Sea Level caused by random and systematic errors. The value is expressed as a linear error at the 90% confidence level.

d. Vertical accuracy, point-to-point. The uncertainty in height between two points caused by random errors. The value is expressed as a linear error at the 90% confidence level.

6.3.2 Attribute. Name of attribute category required by the feature as specified in the Feature Attribute Coding Standard (FACS). Attribute categories are characteristics in menu form relative to a specified feature or features.

6.3.2.1 Attribute code (Acode). Three digit alpha or alpha-numeric character (acronym) FACS code assigned to each attribute category which identifies the attribute category (e.g., ZVL-Z Value). Attribute categories are defined by mutually exclusive sets of attribute values which are feature dependent.

6.3.3 Cell. 1 degree by 1 degree area of coverage.

6.3.4 Circular error. An accuracy figure representing the stated percentage of probability that any point expressed as a function of two linear components (for example, latitude and longitude or northing and easting) will be within the given figure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

6.3.5 DTED. A geographic matrix of terrain elevation data points converted into a numerical format for computer storage and analysis at precise increments of latitude and longitude. The precise increment depends upon the latitude. Elevation values are expressed in meters.

6.3.6 Feature. Name of feature as specified in the FACS. A feature is a physical (e.g., Bridge) or conceptual (e.g., Route - Nautical) entity of the real world which has one or more sets of coordinates to be included on a product.

6.3.6.1 Feature code (Fcode). Five digit alpha-numeric, FACS code assigned to each feature (e.g., 9B045- Elevation Point). The first two digits identify the category and subcategory to which each feature belongs (e.g., 9-General, B-Control Points).

6.3.6.2 Feature type. Designation of feature type. If there is more than one feature type for the feature, then the Acode and inclusion conditions are stated separately for each type.

a. Area - More than two sets of coordinates defining a closed area; areas may span more than one map sheet or geographic area requirement.

b. Line - Two or more coordinate sets defining a series of line segments.

c. Point - One set of coordinates.

6.3.7 Inclusion condition. Conditions under which the feature/attribute(s) are required by the product.

6.3.8 Linear error. Linear error is the difference between the true or known value and the measured or derived value, and is normally expressed in terms of a percentage probability level. For example, LE 90% is the term used to express the linear error at 90% probability, which is the Map Accuracy Standard. This refers to the vertical accuracy of terrain elevations in the digital data base.

6.3.9 Nautical mile. 6076 feet; 1852 meters (international value).

6.3.10 Off-line digital data base. Magnetic media containing information in digital form including header information, geographic coordinates and descriptive information for planimetric features and terrain within a specific geographic area.

6.3.11 Terrain matrix. A rectangular array of elevation values.

6.3.12 WGS (World Geodetic System). A horizontal control datum based upon a framework of marked points or objects on the surface of the earth, the positions of which are commonly known in respect to the earth's center of mass.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the current version before use.

6.4 International standardization agreements.

"Certain provisions of this specification are subject of international standardization agreement. When amendment, revision, or cancellation of this specification is proposed that will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations."

6.4.1 International Standardization Agreements (STANAGs).

This section is not applicable to this specification.

6.4.2 Quadripartite Standardization Agreements (QSTAGs).

This section is not applicable to this specification.

6.4.3 Air Standardization Coordinating Committee Agreements (ASCC AIR STDs/ STDs/ADV PUBs).

This section is not applicable to this specification.

6.4.4 International MC&G agreements.

This section is not applicable to this specification.

6.4.5 Executive orders.

This section is not applicable to this specification.

6.4.6 Inter-Agency agreements.

This section is not applicable to this specification.

6.4.7 Other documentation.

This section is not applicable to this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

INDEX

	PARAGRAPH	PAGE
Absolute horizontal accuracy	3.1.1	2
Absolute vertical accuracy	3.1.2	2
Accuracy	3.1	2
Accuracy (definition)	6.3.1	25
Accuracy description record (ACC)	3.10.5	22
Accuracy subregion description	3.10.5.1	22
Adjacent data files	3.3.1	3
Air Standardization Coordinating Committee Agreements (ASCC AIR STDs/STD's/ADV PUBs	6.4.3	27
Applicable Documents	2.	1
Attribute	6.3.2	25
Attribute code (Acode)	6.3.2.1	25
Cell	6.3.3	25
Circular error	6.3.4	25
Concluding Material		31
DTED	6.3.5	26
Data file structure	3.7.1	4
Data record characteristics	3.8	5
Data record sequence	3.7.5	5
Data records	3.5.2	4
Data set identification (DSI) record	3.10.4	20
Data value sequence	3.7.4	5
Datum	3.2	2
Definitions	6.3	25
Description of record contents	3.9	8
Elevation values	3.8.2	5
Executive orders	6.4.5	27
Explanation of records and fields (DTED)	3.10	19
Feature	6.3.6	26
Feature code (Fcode)	6.3.6.1	26
Feature requirements/inclusion conditions	3.10	23
Feature type	6.3.6.2	26
Fields (DSI)	3.10.4.1	20
Fields (UHL)	3.10.3.1	20
File description	3.6	4
File header label (HDR)	3.10.2	19
File structure diagram	3.7.7	5
Files extent	3.7.2	5
Final product quality	4.1.2	24
Government documents	2.1	1
Hash control total information	3.7.6	5
Horizontal datum	3.2.1	2
Inclusion condition	6.3.7	26
Intended use	6.1	25
Inter-Agency agreements	6.4.6	27
International MC&G agreements	6.4.4	27

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

	PARAGRAPH	PAGE
International standardization agreements	6.4	27
International Standardization Agreements (STANAGs)	6.4.1	27
Linear error	6.3.8	26
Logical characteristics	3.7	4
Missing elevations	3.8.4	8
Nautical mile	6.3.9	26
Non-Government publications	2.2	2
Notes	6.	25
Null elevations	3.8.3	6
Numeric value	3.8.1	5
Off-line digital data base	6.3.10	26
Order of precedence	2.3	2
Other documentation	6.4.7	27
Other Government documents, drawings, and publications	2.1.2	2
Packaging	5.	24
Physical characteristics of magnetic tape	3.4.1	3
Product Format	3.3	3
Purpose	1.2	1
Quadripartite Standardization Agreements (QSTAGs)	6.4.2	27
Quality Assurance Provisions	4.	24
Record characteristics	3.5	3
Record sequence	3.5.3	4
Recorded labels	3.5.1	3
Relative horizontal accuracy (point-to-point)	3.1.3	2
Relative vertical accuracy (point-to-point)	3.1.4	2
Requirements	3.	2
Responsibility for compliance	4.1.1	24
Responsibility for inspection	4.1	24
Scope	1.	1
Scope	1.1	1
Security	1.3	1
Specifications, standards and handbooks	2.1.1	1
Standard format	3.4	3
Supersession	6.2	25
Terrain elevation intervals	3.7.3	5
Terrain matrix	6.1.11	26
Type of Input/Output	3.4.2	3
User header label (UHL)	3.10.3	20
Vertical datum	3.2.2	2
Volume header label (VOL)	3.10.1	19
WGS (World Geodetic System)	6.3.12	26

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
DMA - MP

Review activities:
Air Force - 09
Army - CE
Navy - NO

User activities:
Air Force
Army
Navy

Preparing activity:
DMA - MP

(project MCGT-0003)

☆U.S. GOVERNMENT PRINTING OFFICE: 1990 - 704-034/30474

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-D-89001	26 February 1990

3. DOCUMENT TITLE
Military Specification Digital Terrain Elevation Data (DTED) Level 2 (1CF)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Defense Mapping Agency	(703)285-9199	356-9199
c. ADDRESS (Include Zip Code) ATTN: PRS 8613 Lee Highway, Fairfax, VA 22031-2137	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:54Z
Check the source to verify that this is the currentversion before use.